Stochastic optimization
L-shaped method

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Two-stage linear program with recourse

Suggested reading: Section 3.2, Kall and Wallace

Summary of formulations:

$$\min_{x \in \mathbb{R}^n_+} \left\{ c^T x + E_\xi \left[\min_{y \in \mathbb{R}^p_+} \{ q(\xi)^T y \mid Wy = h(\xi) - T(\xi)x \} \right] \,\middle|\, Ax = b \right\}$$

$$\min_{x \in \mathbb{R}^n_+} \left\{ c^T x + E_\xi \left[v(\xi, h(\xi) - T(\xi)x) \right] \,\middle|\, Ax = b \right\}$$

$$\min_{x \in \mathbb{R}^n_+} \left\{ c^T x + E_\xi \left[Q(x, \xi) \right] \,\middle|\, Ax = b \right\}$$

$$\min_{x \in \mathbb{R}^n_+} \left\{ c^T x + \mathcal{Q}(x) \,\middle|\, Ax = b \right\}$$

We have studied their properties. But how do we solve them?

Notation: reminder

- First-stage feasible set: $K_1 = \{x \in \mathbb{R}^n_+ \mid Ax = b\}$.
- Weak second-stage feasible set: $K_2 = \{x \mid Q(x, \xi) < \infty \text{ a.s.}\}$.
- Strong second-stage feasible set: $K_2^s = \{x \mid \mathcal{Q}(x) < \infty\}$.

These two sets differ in general, but coincide in the framework retained here:

Theorem

For a stochastic program with fixed recourse, where ξ has finite second-order moments, $K_2 = K_2^s$.

We can therefore rewrite the problem as

$$\min_x \{c^T x + \mathcal{Q}(x) \mid x \in K_1 \cap K_2\}.$$

Deterministic equivalent

Assume now that the support of ξ is finite, with S scenarios $\xi_1, \ldots, \xi_S$, scenario s having probability p_s and recourse action y_s. The deterministic equivalent is

$$\min_{x \geq 0, y_1 \geq 0, \ldots, y_S \geq 0} c^T x + \sum_{s=1}^{S} p_s q_s^T y_s$$

s.t.

$$\begin{array}{lllll} Ax & & & & = b \\ T_1 x & + Wy_1 & & & = h_1 \\ T_2 x & & + Wy_2 & & = h_2 \\ \vdots & & & \ddots & \vdots \\ T_S x & & & + Wy_S & = h_S \end{array}$$

A linear program,... but a very large one: $n + pS$ variables and $m_1 + m_2 S$ constraints.

Idea: exploit its structure, and use decomposition techniques.

Decomposition methods

General principle: the nonlinear term (the recourse function) requires computing the solution of all the second-stage recourse linear programs.

Is it possible to avoid repeated evaluations of all the recourse functions in order to compute these solutions?

Idea: build a master program in x, and evaluate the objective function exactly (including all the scenarios) as a subproblem only.

Block structure

Primal:

$$\begin{pmatrix} A & & & & \\ T_1 & W & & & \\ T_2 & & W & & \\ \vdots & & & \ddots & \\ T_S & & & & W \end{pmatrix}$$

Dual:

$$\begin{pmatrix} A^T & T_1^T & T_2^T & \cdots & T_S^T \\ & W^T & & & \\ & & W^T & & \\ & & & \ddots & \\ & & & & W^T \end{pmatrix}$$

These block shapes gave the method its name: L-shaped.

Main ideas

Section 1.6.4, Kall and Wallace

Assume for now that Ξ contains one element only. The problem becomes

$$\begin{aligned} \min_{x,y} \; & c^T x + q^T y \\ \text{s.t. } & Ax = b, \\ & Tx + Wy = h, \\ & x \geq 0, \; y \geq 0. \end{aligned}$$

The problem can be reformulated as

$$\begin{aligned} \min_{x} \; & c^T x + f(x) \\ \text{s.t. } & Ax = b, \; x \geq 0, \end{aligned}$$

with

$$f(x) \stackrel{\text{def}}{=} \min_{y} \{ q^T y \mid Wy = h - Tx, \; y \geq 0 \}.$$

Problem reformulation

Assume $K_1 \cap K_2 \neq \emptyset$, $K_1 = \{x \in \mathbb{R}^n_+ \mid Ax = b\}$ bounded, and the second-stage dual feasible: some $\bar{\pi}$ satisfies $W^T\bar{\pi} \leq q$.
Boundedness of K_1 alone would not suffice — it does not prevent $f(x) = -\infty$. The dual assumption does, by weak duality: $f(x) \geq (h - Tx)^T\bar{\pi}$ for every x, and this affine function is bounded below on the bounded K_1. So f admits a uniform lower bound θ^0 on K_1, and $\inf\{c^Tx + f(x) \mid x \in K_1 \cap K_2\}$ is finite.
Introducing a scalar θ to represent $f(x)$:

$$\begin{aligned} \min_{x,\theta} \; & c^Tx + \theta \\ \text{s.t. } & Ax = b, \\ & \theta \geq f(x), \\ & x \geq 0. \end{aligned}$$

First ideas

Try to construct a sequence of new (additional) linear constraints in order to define a feasible set $\mathcal{B}_1$ of vectors $(x, \theta) \in \mathbb{R}^n \times \mathbb{R}$, such that, with

$$\mathcal{B}_0 = \{(x, \theta) \in \mathbb{R}^n_+ \times \mathbb{R} \mid Ax = b\},$$

the problem

$$\min_{(x,\theta) \in \mathcal{B}_0 \cap \mathcal{B}_1} c^T x + \theta$$

gives a (first-stage) solution of the initial problem and an estimation θ of the second-stage objective value.

Dual decomposition method

Step 1 Let θ^0 be a lower bound for

$$\min_{x,y}\{q^T y \mid Ax = b,\ Tx + Wy = h,\ x \geq 0,\ y \geq 0\},$$

set $\nu := 1$, and let (x^1, θ^1) be a solution of

$$\min_{x,\theta}\{c^T x + \theta \mid Ax = b,\ \theta \geq \theta^0,\ x \geq 0\}.$$

Set also $\mathcal{B}_1 := \{(x, \theta) \in \mathbb{R}^n \times \mathbb{R} \mid \theta \geq \theta^0\}$.

Step 2 Evaluate

$$\begin{aligned} f(x^\nu) &= \min_y\{q^T y \mid Wy = h - Tx^\nu,\ y \geq 0\} \\ &= \max_\pi\{(h - Tx^\nu)^T \pi \mid W^T \pi \leq q\}. \quad \text{(duality)} \end{aligned}$$

(this is the asymmetric dual, recalled in the linear programming background notes)

We have to consider two cases.

Dual decomposition method (cont'd)

Step 2 (a) If $f(x^\nu) = +\infty$, then x^ν is not feasible for the second-stage constraints: no $y \geq 0$ satisfies $Wy = h - Tx^\nu$.

Farkas' lemma (background notes)

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Then, exactly one of the following two statements is true:

- $\exists\, x \in \mathbb{R}^n$ such that $Ax = b$ and $x \geq 0$;
- $\exists\, u \in \mathbb{R}^m$ such that $u^T A \geq 0$ and $u^T b < 0$.

Therefore, if no feasible $y \geq 0$ exists, $\exists\, u$ such that

$$W^T u \geq 0 \text{ and } (h - Tx^\nu)^T u < 0,$$

or, by setting $\sigma^\nu = -u$,

$$W^T \sigma^\nu \leq 0 \text{ and } (h - Tx^\nu)^T \sigma^\nu > 0.$$

Dual decomposition method (cont'd)

Now, given x feasible (for the first and second stages), there exists $y \geq 0$ s.t. $Wy = h - Tx$. For such a y, we have

$$(\sigma^\nu)^T(h - Tx) = (\sigma^\nu)^T Wy \leq 0,$$

since $W^T\sigma^\nu \leq 0$ and $y \geq 0$. Therefore

$$(\sigma^\nu)^T h \leq (\sigma^\nu)^T Tx.$$

This inequality should be valid for any feasible x (therefore excluding x^ν), and we enforce it by adding it to the first-stage linear program: feasibility cut

$$(\sigma^\nu)^T(h - Tx) \leq 0.$$

Redefine $\mathcal{B}_1 := \mathcal{B}_1 \cap \{(x, \theta) \mid (\sigma^\nu)^T(h - Tx) \leq 0\}$.

Dual decomposition method (cont'd)

Step 2 (b) If $f(x^\nu)$ is finite, given x^ν, we have a primal optimal solution y^ν and a dual optimal solution π^ν with

$$f(x^\nu) = (h - Tx^\nu)^T \pi^\nu$$

by strong duality.

Moreover, for any x, we can write

$$\begin{aligned} f(x) &= \sup_\pi \left\{ (h - Tx)^T \pi \mid W^T \pi \leq q \right\} \quad \text{(second-stage dual)} \\ &\geq (h - Tx)^T \pi^\nu \\ &= (\pi^\nu)^T (h - Tx). \end{aligned}$$

So every dual feasible π gives an affine minorant of f. The reformulation requires $\theta \geq f(x)$, hence in particular

$$\theta \geq (\pi^\nu)^T (h - Tx).$$

Dual decomposition method (cont'd)

$\theta \geq (\pi^\nu)^T(h - Tx)$ is violated at (x^ν, θ^ν) iff $(\pi^\nu)^T(h - Tx^\nu) > \theta^\nu$.
In this case we introduce the optimality cut, rejecting the suboptimal solution (x^ν, θ^ν):

$$\theta \geq (\pi^\nu)^T(h - Tx),$$

and we redefine $\mathcal{B}_1 := \mathcal{B}_1 \cap \{(x, \theta) \mid \theta \geq (\pi^\nu)^T(h - Tx)\}$.

If $f(x^\nu) \leq \theta^\nu$, stop: (x^ν, θ^ν) is optimal. Otherwise, go to Step 3.

Step 3 Solve the updated problem

$$\min\{c^T x + \theta \mid (x, \theta) \in \mathcal{B}_0 \cap \mathcal{B}_1\}.$$

Increment ν, and return to Step 2 with the new solution (x^ν, θ^ν).

Computing σ^ν

Farkas' lemma tells us that σ^ν exists when $f(x^\nu) = +\infty$, but not how to find it.

Basic ideas:

- we have detected that the primal is not feasible as the dual is unbounded;
- therefore, there is some feasible direction along which we can increase the dual value as much as we want;
- this direction is the ray σ^ν that we are looking for.

There are many technical details in Birge and Louveaux (Chapter 5); we focus here on the main practical ideas.

Computing σ^ν (cont'd)

Similar to the two-phase method in linear programming. The first phase looks for a feasible solution by solving an artificial program. We do the same, but what we look for is a certificate of infeasibility!

Solve the linear program

$$\begin{aligned} \min\ & w = e^T v^+ + e^T v^- \\ \text{s.t. } & Wy + Iv^+ - Iv^- = h - Tx^\nu \\ & y, v^+, v^- \geq 0, \end{aligned}$$

where $e^T = (1, \ldots, 1)$.

As $v^+, v^- \geq 0$, $w^* \geq 0$:

- if $w^* = 0$, then $v^{+*} = v^{-*} = 0$ and $Wy^* = h - Tx^\nu$; in other words, the second-stage problem is feasible for x^ν;
- if $w^* > 0$, no feasible $y \geq 0$ exists with $Wy = h - Tx^\nu$: such a y would make $(y, 0, 0)$ feasible with $w = 0$, contradicting the optimality of $w^* > 0$.

Computing σ^ν (cont'd)

The dual of the previous problem can be written in two equivalent forms:

$$\begin{aligned} \max_{\sigma} \ & (h - Tx^\nu)^T \sigma \\ \text{s.t. } & W^T \sigma \leq 0 \\ & \sigma \leq e \\ & -\sigma \leq e \end{aligned}$$

$$\begin{aligned} \max_{\sigma} \ & (h - Tx^\nu)^T \sigma \\ \text{s.t. } & W^T \sigma \leq 0 \\ & \|\sigma\|_\infty \leq 1 \end{aligned}$$

By duality, if $w^* > 0$, the optimal solution σ^ν satisfies $(h - Tx^\nu)^T \sigma^\nu > 0$. Moreover, as $W^T \sigma^\nu \leq 0$, σ^ν satisfies the conditions we were looking for!

Convergence: why the stopping test is correct

Stopping rule: $f(x^\nu) \le \theta^\nu$.
Every cut added is valid: it is satisfied by every (x, θ) with $x \in K_1 \cap K_2$ and $\theta \ge f(x)$ — that is how each was derived. So $\mathcal{B}_0 \cap \mathcal{B}_1$ contains the feasible set of the original problem, the master is a relaxation, and

$$c^T x^\nu + \theta^\nu \;=\; \min_{\mathcal{B}_0 \cap \mathcal{B}_1} c^T x + \theta \;\le\; \min_{x \in K_1 \cap K_2} \{c^T x + f(x)\}.$$

Suppose now $f(x^\nu) \le \theta^\nu$. Then $f(x^\nu) < \infty$, so $x^\nu \in K_1 \cap K_2$ and $(x^\nu, f(x^\nu))$ is feasible for the original problem:

$$\min_{x \in K_1 \cap K_2} \{c^T x + f(x)\} \;\le\; c^T x^\nu + f(x^\nu) \;\le\; c^T x^\nu + \theta^\nu.$$

The two chains close on themselves: all the inequalities are equalities, $\theta^\nu = f(x^\nu)$, and x^ν is optimal.

Convergence: why the method stops

Only finitely many distinct cuts can ever be produced.

- An optimality cut is built from a dual *basic* optimal solution, i.e. an extreme point of $\{\pi \mid W^T\pi \leq q\}$ — such points exist when $\operatorname{rank}(W) = m$ (the polyhedron then contains no line), and there are finitely many (background notes).
- A feasibility cut is built from a basic optimal solution σ^ν of the phase-1 dual, i.e. a vertex of the polytope $\{\sigma \mid W^T\sigma \leq 0,\ \|\sigma\|_\infty \leq 1\}$: again finitely many.

And no cut is produced twice: a cut generated at (x^ν, θ^ν) is by construction violated there, whereas (x^ν, θ^ν) satisfies every cut already in $\mathcal{B}_1$. Each non-terminating iteration therefore adds a cut that was not present before, and the stock is finite.

Proposition

If the two-stage stochastic program is solvable and $\{x \mid Ax = b,\ x \geq 0\}$ is bounded, the dual decomposition method gives an optimal solution after a finite number of iterations.

S scenarios

What if Ξ is finite, but not a singleton?

We can introduce feasibility and optimality cuts for all the recourse functions

$$Q_s(x) = Q(x, \xi_s) = \min \left\{ q_s^T y_s \mid W y_s = h_s - T_s x,\ y_s \geq 0 \right\},$$

$s = 1, \ldots, S$, with $q_s = q(\xi_s)$, $h_s = h(\xi_s)$, $T_s = T(\xi_s)$. We obtain a *multi-cut* dual decomposition method.

Alternatively, we can combine cuts corresponding to blocks $s = 1, \ldots, S$ with their respective probabilities.

Single-cut algorithm

With S scenarios, we can proceed as before, replacing $\mathcal{Q}(x)$ with a single scalar θ (single-cut). This leads to the master program

$$\begin{aligned} \min\ & c^T x + \theta \\ \text{s.t. } & Ax = b, \\ & \theta \geq \mathcal{Q}(x), \\ & x \geq 0,\ \theta \in \mathbb{R}. \end{aligned}$$

This program has $n + 1$ variables only!

Master program

Denote by (x^*, θ^*) the optimal solution; we aim to have the constraint on θ active at the optimum:

$$\theta^* = \mathcal{Q}(x^*).$$

How do we achieve this? By using cuts.

A cut is an additional linear constraint, restricting the feasible set of the master program, but preserving the optimal solution of the original problem.

As in the simple example, two types of cuts have to be considered:

- feasibility cut;
- optimality cut.

Feasibility cuts

Solving the master problem can produce a solution (x^ν, θ^ν) such that $\mathcal{Q}(x^\nu) = +\infty$, i.e., $x^\nu \notin K_2$: the second-stage problem is infeasible for at least one scenario.

The feasibility cuts aim to keep the solutions of the master program in $K_1 \cap K_2$. If our problem is well defined, we should have $K_1 \cap K_2 \neq \emptyset$. For any feasible $x \in K_1 \cap K_2$, there exists $y_s \geq 0$ such that $Wy_s = h_s - T_s x$ for all $s = 1, \ldots, S$.

We proceed as with a single scenario, but what we look for is an infeasible scenario! For $s = 1, \ldots, S$, solve the phase-1 program with right-hand side $h_s - T_s x^\nu$, until $w^* > 0$ for some scenario s. Its dual then gives σ^ν, and the feasibility cut

$$(\sigma^\nu)^T (h_s - T_s x) \leq 0.$$

Example: Birge and Louveaux (p. 174)

Consider the problem

$$
\begin{aligned}
\min_x \; & 3x_1 + 2x_2 + E_{\boldsymbol{\xi}}[\min_y -15y_1 - 12y_2] \\
\text{s.t. } & 3y_1 + 2y_2 \leq x_1 \\
& 2y_1 + 5y_2 \leq x_2 \\
& 0.8\xi_1 \leq y_1 \leq \xi_1 \\
& 0.8\xi_2 \leq y_2 \leq \xi_2 \\
& x \geq 0, \; y \geq 0
\end{aligned}
$$

The random vector $\boldsymbol{\xi}$ takes values in

$$
\boldsymbol{\xi} \in \{(4,6),\, (4,8),\, (6,6),\, (6,8)\}, \quad \text{each with probability } p_s = 0.25.
$$

Deterministic equivalent

Interpretation: we face two investment decisions x_1 and x_2, that must allow at least 80% of the demand to be satisfied.
Deterministic equivalent:

$$\begin{aligned}
\min_{x\geq 0, y\geq 0} \; & 3x_1 + 2x_2 - \sum_{s=1}^{4} \frac{1}{4}\left(15y_1^{(s)} + 12y_2^{(s)}\right) \\
\text{s.t. } & 3y_1^{(s)} + 2y_2^{(s)} \leq x_1, \; s = 1,\dots,4 \\
& 2y_1^{(s)} + 5y_2^{(s)} \leq x_2, \; s = 1,\dots,4 \\
& 0.8\xi_1^{(s)} \leq y_1^{(s)} \leq \xi_1^{(s)}, \; s = 1,\dots,4 \\
& 0.8\xi_2^{(s)} \leq y_2^{(s)} \leq \xi_2^{(s)}, \; s = 1,\dots,4 \\
& x \geq 0, \; y^{(s)} \geq 0, \; s = 1,\dots,4
\end{aligned}$$

where $\xi^{(1)} = (4,6)$, $\xi^{(2)} = (4,8)$, $\xi^{(3)} = (6,6)$, $\xi^{(4)} = (6,8)$.

Deterministic equivalent: optimal solution

It is possible to solve the deterministic equivalent.

We obtain the solution

$$
\begin{aligned}
(x_1^*, x_2^*) &= (27.2, 41.6) \\
\left(y_1^{(1)*}, y_2^{(1)*}\right) &= (4, 6) \\
\left(y_1^{(2)*}, y_2^{(2)*}\right) &= (4, 6.72) \\
\left(y_1^{(3)*}, y_2^{(3)*}\right) &= (5.0667, 6) \\
\left(y_1^{(4)*}, y_2^{(4)*}\right) &= (4.8, 6.4)
\end{aligned}
$$

with an optimal objective value of 22.44.
Can we limit capacity expansion?

Example B&L: master program

We now build the master program

$$\min_{x,\theta} 3x_1 + 2x_2 + \theta$$
$$\text{s.t. } x \geq 0.$$

Since there is currently no constraint on θ, we simply ignore it. The solution is then $x^1 = (0, 0)$, at which no scenario is feasible!

In order to keep the discussion short, we will consider scenario $s = 4$, with $\xi^{(4)} = (6, 8)$ (in general, it is impossible to automatically choose the most convenient scenario).

Feasibility cut

In order to determine the feasibility cut, we solve the program

$$\begin{aligned}
\min_{y,v} w = & v_1^- + v_2^- + v_3^+ + v_4^- + v_5^+ + v_6^- \\
\text{s.t. } & 3y_1 + 2y_2 + v_1^+ - v_1^- = 0 \\
& 2y_1 + 5y_2 + v_2^+ - v_2^- = 0 \\
& y_1 + v_3^+ - v_3^- = 4.8 \\
& y_1 + v_4^+ - v_4^- = 6.0 \\
& y_2 + v_5^+ - v_5^- = 6.4 \\
& y_2 + v_6^+ - v_6^- = 8.0 \\
& y \geq 0, \ v \geq 0.
\end{aligned}$$

The optimal value is $w^* = 11.2$, associated with $v_3^+ = 4.8$ and $v_5^+ = 6.4$, and the dual vector (obtained with GLPK)

$$\sigma^1 = \begin{pmatrix} -1 & -1 & 1 & 0 & 1 & 0 \end{pmatrix}^T$$

Feasibility cut (cont'd)

We define the feasibility cut

$$D_1 x \geq d_1,$$

with

$$D_1 = (\sigma^1)^T T = \begin{pmatrix} -1 & -1 & 1 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \end{pmatrix}$$

and

$$d_1 = (\sigma^1)^T h^{(4)} = \begin{pmatrix} -1 & -1 & 1 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 4.8 \\ 6.0 \\ 6.4 \\ 8.0 \end{pmatrix} = 11.2$$

Feasibility cut (cont'd)

We obtain the cut

$$x_1 + x_2 \geq 11.2$$

which excludes $x^1 = (0,\ 0)$.

Birge and Louveaux report the cut

$$3x_1 + x_2 \geq 123.2$$

Can we justify this difference?

The dual solution is not unique, and we can also choose

$$\sigma^1 = \begin{pmatrix} -3/11 & -1/11 & 1 & 0 & 1 & 0 \end{pmatrix}^T.$$

d_1 does not change, but D_1 becomes $(3/11\ 1/11)$. This solution makes the first two dual constraints active. In other words, this solution corresponds to a different dual basis.

Dual

$$
\begin{aligned}
\max_{\sigma} \; & 4.8\sigma_3 + 6.0\sigma_4 + 6.4\sigma_5 + 8.0\sigma_6 \\
\text{s.t. } & 3\sigma_1 + 2\sigma_2 + \sigma_3 + \sigma_4 \leq 0 \quad (y_1) \\
& 2\sigma_1 + 5\sigma_2 + \sigma_5 + \sigma_6 \leq 0 \quad (y_2) \\
& \sigma_1 \leq 0 \quad (v_1^+), \; -\sigma_1 \leq 1 \quad (v_1^-) \\
& \sigma_2 \leq 0 \quad (v_2^+), \; -\sigma_2 \leq 1 \quad (v_2^-) \\
& \sigma_3 \leq 1 \quad (v_3^+), \; -\sigma_3 \leq 0 \quad (v_3^-) \\
& \sigma_4 \leq 0 \quad (v_4^+), \; -\sigma_4 \leq 1 \quad (v_4^-) \\
& \sigma_5 \leq 1 \quad (v_5^+), \; -\sigma_5 \leq 0 \quad (v_5^-) \\
& \sigma_6 \leq 0 \quad (v_6^+), \; -\sigma_6 \leq 1 \quad (v_6^-)
\end{aligned}
$$

Dual (cont'd)

By the complementarity conditions, $v_3^+ > 0$ and $v_5^+ > 0$ imply that $\sigma_3 = 1$ and $\sigma_5 = 1$. Strong duality requires that $\sigma_4 = \sigma_6 = 0$ in order to have the dual objective value 11.2.

The optimal solutions can however differ on σ_1 and σ_2.

The solution

$$\begin{pmatrix} -1 & -1 & 1 & 0 & 1 & 0 \end{pmatrix}$$

sets two bound constraints active for σ_1 and σ_2, while the solution obtained by Birge and Louveaux makes these constraints inactive, with the first two dual constraints active instead.

Feasibility cuts: B&L results

Using CPLEX or Gurobi produces the B&L cut, and the first-stage solution becomes $x^2 = (41.067, 0)^T$.

If we continue as in B&L, a second feasibility cut is generated: $x_2 \geq 22.4$. The first-stage solution becomes $x^3 = (33.6, 22.4)^T$.

A third feasibility cut is generated: $x_2 \geq 41.6$. The first-stage decision is now $x^4 = (27.2, 41.6)^T \in K_1 \cap K_2$.

Another choice for the initial realization would have produced different cuts.

On the problem formulation

The production of feasibility cuts can be inefficient.

With $\xi^{(4)} = (6, 8)$, it is easy to see that we must have $y_1 \geq 4.8$ and $y_2 \geq 6.4$.

From the first two second-stage constraints, this implies $x_1 \geq 27.2$ and $x_2 \geq 41.6$.

We could therefore consider the following initial program

$$\begin{aligned} \min\ & 3x_1 + 2x_2 + \mathcal{Q}(x) \\ \text{s.t. } & x_1 \geq 27.2, \\ & x_2 \geq 41.6. \end{aligned}$$

The recourse is now relatively complete!

Optimality cuts

The problem

$$\begin{aligned} &\min\ c^T x + \mathcal{Q}(x) \\ &\text{s.t. } x \in K_1 \cap K_2, \end{aligned}$$

is equivalent to

$$\begin{aligned} &\min\ c^T x + \theta \\ &\text{s.t. } \theta \geq \mathcal{Q}(x), \\ &\qquad x \in K_1 \cap K_2. \end{aligned}$$

The constraint $\theta \geq \mathcal{Q}(x)$ cannot be directly used, and will be expressed by means of optimality cuts.

Master problem

Initially, the only constraint on θ is $\theta \in \mathbb{R}$, and we obtain the infimum of the objective with $\theta = -\infty$.

As long as $\theta = -\infty$, we remove it from the objective function. In other words, we solve the first-stage problem only.

Temporarily ignore this constraint. We obtain an optimal solution (x^ν, θ^ν), with θ^ν set to $-\infty$ during the first iteration of the method.

The structure of the deterministic equivalent is exploited by minimizing each second-stage function separately, and the decomposition is based on the primal problem. Therefore, this is a primal decomposition method, also known as Benders decomposition.

Optimality cut

Consider the expected second-stage cost

$$\mathcal{Q}(x) = \sum_{s=1}^{S} p_s Q(x, \xi_s),$$

with

$$Q(x, \xi_s) = \min\{q_s^T y \mid Wy = h_s - T_s x,\ y \geq 0\}.$$

We know that $Q(x, \xi_s)$ is convex and piecewise linear, and since Ξ is finite, $\mathcal{Q}$ is also convex and piecewise linear.

Optimality cut (cont'd)

For $x \in K_2$, the second-stage problem is feasible for every ξ; its dual is

$$\max\{\pi^T(h(\xi) - T(\xi)x) \mid W^T\pi \leq q(\xi)\}.$$

- The dual feasible set $\{\pi \mid W^T\pi \leq q(\xi)\}$ does not involve x, as the recourse matrix W is fixed: given ξ, the dual is either feasible for every x, or for none. If q does not depend on ξ, the dual is feasible for every (x, ξ), or for none.
- As the primal is feasible, only two primal–dual situations remain (background notes): either the dual is feasible, and both problems have optimal solutions with equal values, or the dual is infeasible, and the primal is unbounded.
- Hence $Q(x, \xi_s)$ is either finite on all of K_2, or $-\infty$ on all of K_2. We assume the first case: $\{\pi \mid W^T\pi \leq q_s\} \neq \emptyset$ for every s, a condition on the data only, as for a single scenario.

Subproblems

Consider the master problem solution (x^ν, θ^ν). We have

$$\mathcal{Q}(x^\nu) = \sum_{s=1}^{S} p_s Q(x^\nu, \xi_s) = \sum_{s=1}^{S} p_s q_s^T y_s^\nu,$$

where y_s^ν is the second-stage optimal solution associated with ξ_s.

By strong duality, we also have

$$\sum_{s=1}^{S} p_s q_s^T y_s^\nu = \sum_{s=1}^{S} p_s (\pi_s^\nu)^T (h_s - T_s x^\nu),$$

where π_s^ν is the dual optimal solution associated with ξ_s.

The constraints of the s-th dual problem, $W^T \pi \leq q_s$, do not involve x.

Subproblems (cont'd)

For any $x \in K_1 \cap K_2$, taking the S duals together,

$$\mathcal{Q}(x) = \max_{\pi_1,\ldots,\pi_S} \left\{ \sum_{s=1}^{S} p_s \pi_s^T (h_s - T_s x) \;\middle|\; W^T \pi_s \le q_s,\ \forall s \right\}$$
$$\ge \sum_{s=1}^{S} p_s (\pi_s^\nu)^T (h_s - T_s x),$$

each π_s^ν being feasible for its own dual, whose constraints do not involve x.
The constraint $\theta \ge \mathcal{Q}(x)$ implies

$$\theta \ge \sum_{s=1}^{S} p_s (\pi_s^\nu)^T (h_s - T_s x) = \alpha + \beta^T x,$$

or

$$-\beta^T x + \theta \ge \alpha,$$

with $\alpha = \sum_{s=1}^{S} p_s (\pi_s^\nu)^T h_s$ and $\beta = -\sum_{s=1}^{S} p_s T_s^T \pi_s^\nu$.

Master problem

Taking account of feasibility and optimality cuts, the master problem can be formulated as:

$$
\begin{aligned}
\min\ & c^T x + \theta \\
\text{s.t. } & Ax = b, \\
& D_\ell x \geq d_\ell,\ \ell = 1, \ldots, K, && \text{(feasibility cuts)} \\
& -\beta_\ell^T x + \theta \geq \alpha_\ell,\ \ell = 1, \ldots, L, && \text{(optimality cuts)} \\
& x \geq 0,\ \theta \in \mathbb{R}.
\end{aligned}
$$

Let (x^ν, θ^ν) be an optimal solution, with ν the iteration index. If no optimality cut is present, θ^ν is set to $-\infty$ and is not considered in the computation of x^ν.

L-shaped algorithm

Step 0. Initialization: $\nu := 0$, $K := 0$, $L := 0$, $\theta^0 := -\infty$.

Step 1. Increment ν and solve the master problem in order to get (x^ν, θ^ν). Stop if the program is not feasible (for $\nu = 1$: $K_1 = \emptyset$).

Step 2. Generate a feasibility cut. For $s = 1, \ldots, S$, solve the linear problem

$$\begin{aligned} \min\ & w = e^T v^+ + e^T v^- \\ \text{s.t. } & Wy + Iv^+ - Iv^- = h_s - T_s x^\nu \\ & y, v^+, v^- \geq 0, \end{aligned}$$

where $e^T = (1, \ldots, 1)$, until $w^* > 0$ for some scenario s.

L-shaped algorithm (cont'd)

Step 2. (continuation) Let σ^ν be the optimal solution of the dual program, and define

$$D_{K+1} = (\sigma^\nu)^T T_s,$$
$$d_{K+1} = (\sigma^\nu)^T h_s,$$

in order to generate the feasibility cut $D_{K+1}x \geq d_{K+1}$. Increment K and return to Step 1.

If $w^* = 0$ for all $s = 1, \ldots, S$, go to Step 3.

L-shaped algorithm (cont'd)

Step 3. Compute $\mathcal{Q}(x^\nu)$, by solving each second-stage problem: for $s = 1, \ldots, S$, solve

$$\begin{aligned} \min\ & q_s^T y \\ \text{s.t. } & Wy = h_s - T_s x^\nu, \\ & y \geq 0. \end{aligned}$$

Stop if $\theta^\nu \geq \mathcal{Q}(x^\nu)$: x^ν is an optimal solution.
Otherwise, increment L, build the optimality cut $-\beta_L^T x + \theta \geq \alpha_L$, and return to Step 1.

Remarks

A pseudo-code is also available in Kall and Wallace.

Theorem

When ξ has finite support, the L-shaped algorithm converges in a finite number of iterations towards an optimal solution if one exists, or establishes that the two-stage (linear, with fixed recourse) stochastic program is infeasible.

Subgradients and optimality cuts

$\mathcal{Q}$ is convex and piecewise linear: at the kinks, subgradients replace gradients.

An optimal dual solution of a linear program is a subgradient of its value function with respect to the right-hand side (background notes). For $x^\nu \in K_2$, as the right-hand side $h_s - T_s x$ is affine in x,

$$-T_s^T \pi_s^\nu \in \partial Q(x^\nu, \xi_s), \qquad \text{hence} \qquad \beta = -\sum_{s=1}^{S} p_s T_s^T \pi_s^\nu \in \partial \mathcal{Q}(x^\nu).$$

As $\alpha + \beta^T x^\nu = \mathcal{Q}(x^\nu)$, the boundary of the optimality cut, $\theta = \mathcal{Q}(x^\nu) + \beta^T(x - x^\nu)$, is a supporting hyperplane to the epigraph of $\mathcal{Q}$ at $(x^\nu, \mathcal{Q}(x^\nu))$. Together, the optimality cuts build an outer approximation of the epigraph of $\mathcal{Q}$, and the feasibility cuts one of K_2.

Optimality cuts: example

Inspired by Birge and Louveaux

We consider the problem

$$\min_x \; c^T x + E_\xi[Q(x,\xi)],$$
$$\text{s.t. } 0 \leq x \leq 10.$$

Assume that $c = 0$, so that the program can be reduced to the second stage, with a second-stage problem defined as

$$\min y^+ + y^- \text{ s.t. } y^+ - y^- = \xi - x,\ y^+ \geq 0,\ y^- \geq 0.$$

It is possible to analytically compute $Q(x,\xi)$, that can be expressed as:

$$Q(x,\xi) = \begin{cases} \xi - x, & \text{if } x \leq \xi, \\ x - \xi, & \text{if } x \geq \xi. \end{cases}$$

$\mathcal{Q}(x)$

What about $\mathcal{Q}(x)$?

Assume that ξ takes the values 1, 2 and 4 with probabilities $p_1 = p_2 = p_3 = 1/3$.

A previous analysis showed that we have to examine 4 cases:

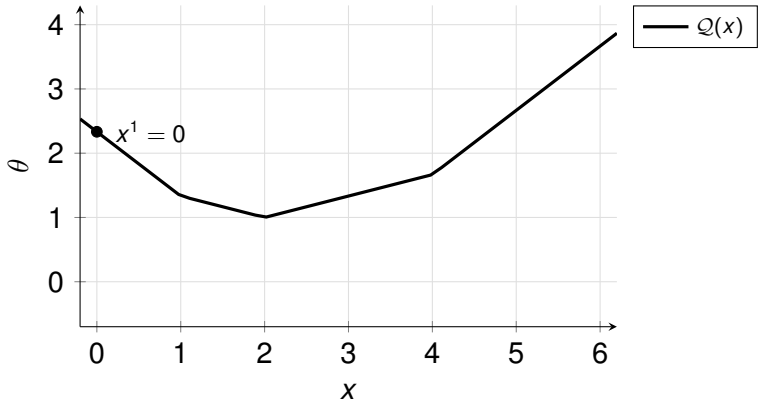
1. $x \leq 1$: $\mathcal{Q}(x) = 7/3 - x$;
2. $1 \leq x \leq 2$: $\mathcal{Q}(x) = 5/3 - x/3$;
3. $2 \leq x \leq 4$: $\mathcal{Q}(x) = x/3 + 1/3$;
4. $4 \leq x$: $\mathcal{Q}(x) = x - 7/3$;

Application of the *L*-shaped algorithm

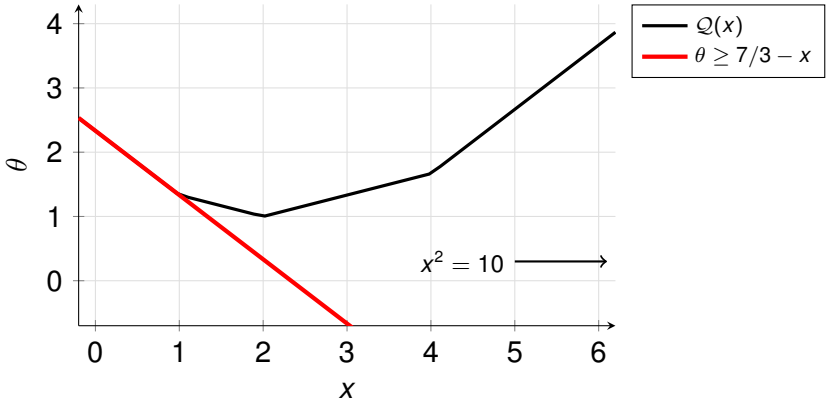
We assume that the starting point is $x^1 = 0$. The sequence of iterates for the *L*-shaped method is as follows.

Iteration 1 x^1 is not optimal; create the cut $\theta \geq 7/3 - x$.
Iteration 2 $x^2 = 10$, $\theta^2 = -23/3$ are not optimal; create the cut $\theta \geq x - 7/3$.
Iteration 3 $x^3 = 7/3$, $\theta^3 = 0$ are not optimal; create the cut $\theta \geq (x + 1)/3$.
Iteration 4 $x^4 = 1.5$, $\theta^4 = 2.5/3$ are not optimal; create the cut $\theta \geq (5 - x)/3$.
Iteration 5 $x^5 = 2$, $\theta^5 = 1$, which is the optimal solution.

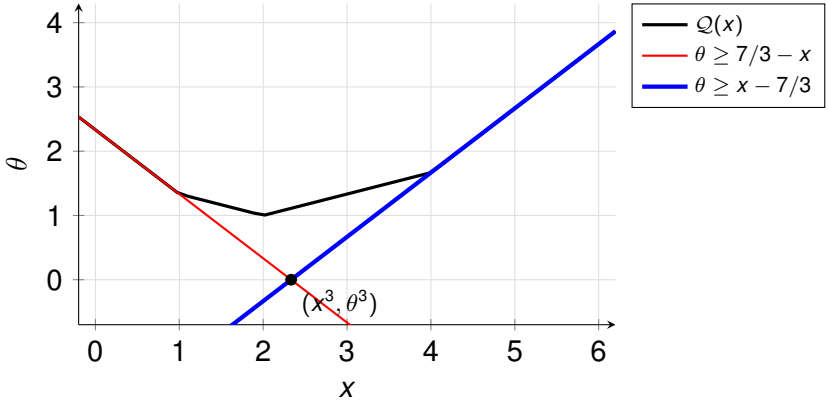
$\mathcal{Q}(x)$, no cut



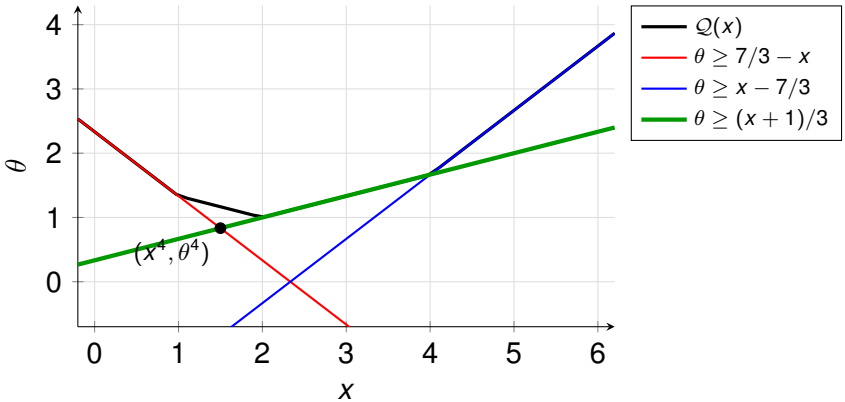
$\mathcal{Q}(x)$, addition of the cut $\theta \geq 7/3 - x$



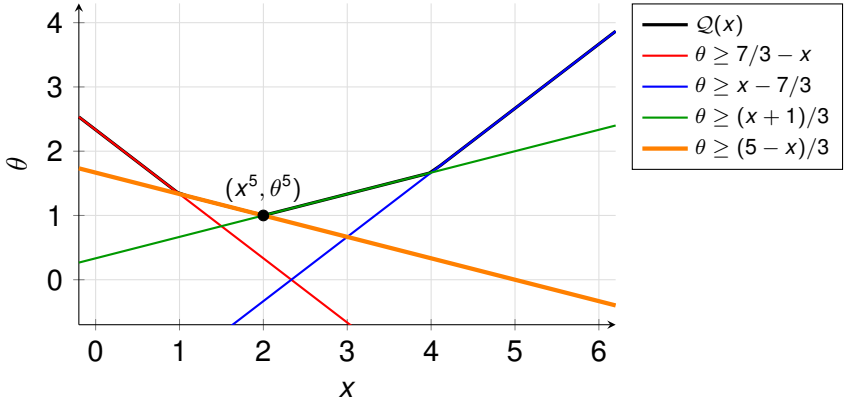
$\mathcal{Q}(x)$, addition of the cut $\theta \geq x - 7/3$



$\mathcal{Q}(x)$, addition of the cut $\theta \geq (x+1)/3$



$\mathcal{Q}(x)$, addition of the cut $\theta \geq (5 - x)/3$



Multi-cut version

- Why do we build one cut only? Idea: approximate each $p_s Q(x, \xi_s)$ by an auxiliary variable θ_s, $s = 1, \ldots, S$.
- We now introduce a cut by scenario

$$\theta_s \geq p_s(\pi_s^\nu)^T(h_s - T_s x) = \alpha_s + \beta_s^T x, \ s = 1, \ldots, S.$$

  If no cut can be built, the solution is optimal.
- Which method works best?
  - Multi-cut: more detailed information, so in general, we can expect fewer major iterations, but as more cuts are added, the first-stage program is more expensive to solve.
  - Rule of thumb: the multi-cut approach is more efficient when the number of realizations S is not significantly greater than the number of first-stage constraints.

Multicut L-shaped method

Step 0 With θ_s^0 a lower bound for $p_s Q(x, \xi_s)$, define

$$\mathcal{B}_0 = \{(x, \theta_1, \ldots, \theta_S) \in \mathbb{R}^n_+ \times \mathbb{R}^S \mid Ax = b\},$$
$$\mathcal{B}_1 = \{(x, \theta_1, \ldots, \theta_S) \in \mathbb{R}^n_+ \times \mathbb{R}^S \mid \theta_s \geq \theta_s^0,\ s = 1, \ldots, S\}.$$

Set $\nu = 0$.

Step 1 Increment ν. Solve the master problem

$$\min \left\{ c^T x + \sum_{s=1}^{S} \theta_s \;\middle|\; (x, \theta_1, \theta_2, \ldots, \theta_S) \in \mathcal{B}_0 \cap \mathcal{B}_1 \right\}.$$

Record the solution $(x^\nu, \theta_1^\nu, \theta_2^\nu, \ldots, \theta_S^\nu)$.

Step 2 Evaluate $\mathcal{Q}(x^\nu) = \sum_{s=1}^{S} p_s Q(x^\nu, \xi_s)$.
$\mathcal{Q}(x^\nu) < \infty$?

Multicut L-shaped method: step 2

- If $\mathcal{Q}(x^\nu) = \infty$, there is a realization ξ_s for which $Q(x^\nu, \xi_s) = \infty$, we add a feasibility cut and set

$$\mathcal{B}_1 := \mathcal{B}_1 \cap \{(x, \theta_1, \theta_2, \ldots, \theta_S) \mid (\sigma^\nu)^T(h_s - T_s x) \leq 0\}.$$

  Note: this inequality is the same as in the single-cut version.
  Return to Step 1.
- If $\mathcal{Q}(x^\nu) < \infty$, we are able to solve all the second-stage problems, for all the scenarios, and obtain the corresponding dual solutions π_s^ν. We get the subgradients

$$u_s = -T_s^T \pi_s^\nu \in \partial Q(x^\nu, \xi_s).$$

Multicut L-shaped method: step 2 - $\mathcal{Q}(x^\nu) < \infty$

- If $p_s Q(x^\nu, \xi_s) \leq \theta_s^\nu$, $s = 1, \ldots, S$, stop: x^ν is optimal.
- If $p_s Q(x^\nu, \xi_s) > \theta_s^\nu$,

$$\mathcal{B}_1 := \mathcal{B}_1 \cap \{(x, \theta_1, \theta_2, \ldots, \theta_S) \mid \theta_s \geq p_s Q(x^\nu, \xi_s) + p_s u_s^T (x - x^\nu)\}.$$

Return to Step 1.

Hybrid approaches

- Partition the scenarios into C "clusters" $\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_C$, and define

$$\mathcal{Q}_{[\mathcal{S}_k]}(x) = \sum_{s \in \mathcal{S}_k} p_s Q(x, \xi_s).$$

  Thus

$$\mathcal{Q}(x) = \sum_{k=1}^{C} \mathcal{Q}_{[\mathcal{S}_k]}(x)$$

  and

$$-\sum_{s \in \mathcal{S}_k} p_s T_s^T \pi_s^\nu \in \partial \mathcal{Q}_{[\mathcal{S}_k]}(x^\nu).$$

- We can therefore approximate each $\mathcal{Q}_{[\mathcal{S}_k]}(x)$ with a cut based on the subgradient inequality.

Algorithm

Step 0 Let θ_k^0 be a lower bound for $\mathcal{Q}_{[\mathcal{S}_k]}(x)$. Set

$$\mathcal{B}_0 = \{(x, \theta_1, \ldots, \theta_C) \in \mathbb{R}^n_+ \times \mathbb{R}^C \mid Ax = b\},$$
$$\mathcal{B}_1 = \{(x, \theta_1, \ldots, \theta_C) \in \mathbb{R}^n_+ \times \mathbb{R}^C \mid \theta_k \geq \theta_k^0,\ k = 1, \ldots, C\}.$$

Set $\nu = 0$.

Step 1 Increment ν and solve the master problem

$$\min c^T x + \sum_{k=1}^{C} \theta_k \quad \text{s.t. } (x, \theta_1, \theta_2, \ldots, \theta_C) \in \mathcal{B}_0 \cap \mathcal{B}_1.$$

Let $(x^\nu, \theta_1^\nu, \theta_2^\nu, \ldots, \theta_C^\nu)$ be the solution found.

Algorithm - Step 2

- Evaluate $\mathcal{Q}(x^\nu) = \sum_{s=1}^{S} p_s Q(x^\nu, \xi_s)$.
- If $\mathcal{Q}(x^\nu) = \infty$, there is some ξ_s such that $Q(x^\nu, \xi_s) = \infty$. We add a feasibility cut:

$$\mathcal{B}_1 := \mathcal{B}_1 \cap \left\{ (x, \theta_1, \ldots, \theta_C) \mid (\sigma^\nu)^T (h_s - T_s x) \leq 0 \right\}.$$

  This inequality has no term in $\theta_1, \ldots, \theta_C$; it is the same inequality as in the standard L-shaped method.
  Go back to Step 1.
- If $\mathcal{Q}(x^\nu) < \infty$, the S LPs corresponding to the S scenarios can be solved, with dual solutions π_s^ν, and

$$u_k := -\sum_{s \in \mathcal{S}_k} p_s T_s^T \pi_s^\nu \in \partial \mathcal{Q}_{[\mathcal{S}_k]}(x^\nu).$$

  If $\mathcal{Q}_{[\mathcal{S}_k]}(x^\nu) \leq \theta_k^\nu$, $k = 1, \ldots, C$, stop: x^ν is optimal.

Algorithm - Step 2 (cont'd)

For each k such that $\mathcal{Q}_{[\mathcal{S}_k]}(x^\nu) > \theta_k^\nu$, add an optimality cut:

$$\mathcal{B}_1 := \mathcal{B}_1 \cap \left\{ (x, \theta_1, \theta_2, \ldots, \theta_C) \mid \theta_k \geq \mathcal{Q}_{[\mathcal{S}_k]}(x^\nu) + u_k^T (x - x^\nu) \right\}.$$

Example

Birge and Louveaux, Section 5.1, Exercise 5, page 200

Assume $n = 1$ (dimension of x), $m_1 = 0$ (no first-stage constraint), $m_2 = 3$ (three second-stage constraints), $p = 6$ (dimension of y), and

$$W = \begin{pmatrix} 1 & -1 & -1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

Assume moreover $S = 2$, and that the two realizations of ξ have the same probability. These realizations are

$$\xi_1 = (q_1, h_1, T_1)^T,$$
$$\xi_2 = (q_2, h_2, T_2)^T.$$

Example (cont'd)

Assume

$$q_1 = (1,0,0,0,0,0)^T,\ q_2 = (3/2,0,2/7,1,0,0)^T,$$
$$h_1 = (-1,2,7)^T,\ h_2 = (0,2,7)^T,$$
$$T_1 = T_2 = (1,0,0)^T.$$

For the first value ξ_1, we have thus the second-stage program

$$\begin{aligned} \min_y\ & y_1 \\ \text{s.t. } & y_1 - y_2 - y_3 - y_4 = -1 - x, \\ & y_2 + y_5 = 2, \\ & y_3 + y_6 = 7, \\ & y \geq 0. \end{aligned}$$

Example (cont'd)

It is easy to see that, for ξ_1,

$$Q_1(x) = \begin{cases} -x - 1 & \text{if } x \leq -1, \\ 0 & \text{if } x \geq -1 \end{cases}$$

For ξ_2, we can show that

$$Q_2(x) = \begin{cases} -1.5x & \text{if } x \leq 0, \\ 0 & \text{if } 0 \leq x \leq 2, \\ \frac{2}{7}(x-2) & \text{if } 2 \leq x \leq 9, \\ x - 7 & \text{if } x \geq 9. \end{cases}$$

Assume also that $-20 \leq x \leq 20$ and $c = 0$.

What are the (expected) properties that we have for $Q_1(x)$ and $Q_2(x)$?
What about the recourse?

Example: *L*-shaped

Start from $x = -2$. We have the following sequence of iterates

Iteration 1 $x^1 = -2$, θ^1 is omitted and we add the cut

$$\theta \geq -0.5 - 1.25x.$$

Iteration 2 $x^2 = 20$, $\theta^2 = -25.5$, new cut:

$$\theta \geq 0.5x - 3.5.$$

Iteration 3 $x^3 = 12/7$, $\theta^3 = -37/14$, new cut:

$$\theta \geq 0.$$

Iteration 4 $x^4 \in [-2/5, 7]$, $\theta^4 = 0$. If x^4 is chosen as any value in $[0, 2]$, stop.

Example: multicut *L*-shaped

The multicut approach would give the following sequence.

- Iteration 1 $x^1 = -2$, θ_1^1 and θ_2^1 are omitted; we add the cuts $\theta_1 \geq -0.5 - 0.5x$, $\theta_2 \geq -3/4x$. Note: $\theta_1 \geq \frac{1}{2}(-x - 1)$ and $\theta_2 \geq \frac{1}{2}(-1.5x)$
- Iteration 2 $x^2 = 20$, $\theta_1^2 = -10.5$, $\theta_2^2 = -15$; new cuts: $\theta_1 \geq 0$, $\theta_2 \geq 0.5x - 3.5$.
- Iteration 3 $x^3 = 2.8$, $\theta_1^3 = 0$, $\theta_2^3 = -2.1$; new cut: $\theta_2 \geq \frac{1}{7}(x - 2)$.
- Iteration 4 $x^4 = 0.32$, $\theta_1^4 = 0$, $\theta_2^4 = -0.24$; new cut: $\theta_2 \geq 0$.
- Iteration 5 $x^5 = 0$, $\theta_1^5 = \theta_2^5 = 0$. Stop.

This example shows that the multicut approach can sometimes take more iterations than the single-cut version.

Computing the EVPI and the VSS

Recall the four values of the previous chapter, with $z(x, \xi) = c^T x + Q(x, \xi)$ and $\overline{\xi} = \sum_{s=1}^{S} p_s \xi_s$:

$$\text{RP} = \min_{x \in K_1 \cap K_2} c^T x + \mathcal{Q}(x), \qquad \text{WS} = \sum_{s=1}^{S} p_s \min_{x \in K_1 \cap K_2(\xi_s)} z(x, \xi_s),$$

$$\text{EV} = \min_{x \in K_1 \cap K_2(\overline{\xi})} z(x, \overline{\xi}), \qquad \text{EEV} = c^T \overline{x} + \mathcal{Q}(\overline{x}),$$

where $\overline{x} = \overline{x}(\overline{\xi})$ is an optimal solution of the EV problem, and

$$\text{EVPI} = \text{RP} - \text{WS}, \qquad \text{VSS} = \text{EEV} - \text{RP}.$$

Question: what does each of them cost to compute, with the tools of this chapter?

RP and WS

- RP is the optimal value of the two-stage program: one run of the L-shaped method, each iteration solving the master problem and the S subproblems.
- WS asks for S independent problems, one per scenario, in which x may depend on ξ_s:

$$\min_{x,y}\{c^T x + q_s^T y \mid Ax = b,\ T_s x + Wy = h_s,\ x \geq 0,\ y \geq 0\}.$$

  Each is a deterministic program with $n_1 + n_2$ variables — or the L-shaped method with a single scenario.
- The S problems can be solved in parallel: they have $S(n_1 + n_2)$ variables in all, against $n_1 + Sn_2$ for the extensive form of RP, but share nothing.
- Their solutions x_s^* show how much the first-stage decision would change with the information.

EV and EEV

- EV is a single deterministic program, at $\overline{\xi}$. This requires T, h and q to be defined at $\overline{\xi}$, which need not be one of the scenarios.
- EEV involves no optimization over x: $\overline{x}$ is fixed, and

$$\text{EEV} = c^T \overline{x} + \sum_{s=1}^{S} p_s Q(\overline{x}, \xi_s)$$

  is exactly what step 2 of the L-shaped method computes at $x^\nu = \overline{x}$: the S subproblems.
- If one of the subproblems is infeasible, $\overline{x} \notin K_2$ and $\text{EEV} = \text{VSS} = +\infty$. The L-shaped method would add a feasibility cut at this point.

Using the *L*-shaped iterations

- Evaluating the subproblems at $\overline{x}$ before the first master problem gives EEV for free, together with a first optimality or feasibility cut, at a well-informed point.
- At iteration ν, once θ is bounded by a cut, the master problem gives a lower bound and the best point visited an upper bound:

$$L^\nu = c^T x^\nu + \theta^\nu \leq \text{RP} \leq U^\nu = \min_{k \leq \nu} c^T x^k + Q(x^k).$$

- Before convergence, the two measures are therefore already bracketed:

$$L^\nu - \text{WS} \leq \text{EVPI} \leq U^\nu - \text{WS}, \qquad \text{EEV} - U^\nu \leq \text{VSS} \leq \text{EEV} - L^\nu.$$

- The method can be stopped as soon as these intervals answer the question asked, for instance whether VSS is large enough to justify the stochastic model.

Example: the farmer

Back to the farmer of the first chapter (Birge and Louveaux, Section 1.1), in profits, so that $\mathrm{EVPI} = \mathrm{WS} - \mathrm{RP}$ and $\mathrm{VSS} = \mathrm{RP} - \mathrm{EEV}$.

	wheat, corn, beets (acres)	profit
WS, above-average yields	183.33, 66.67, 250	167 667
WS, average yields	120, 80, 300	118 600
WS, below-average yields	100, 25, 375	59 950
WS		115 406
RP	170, 80, 250	108 390
EV, at the average yields	$\overline{x} = (120, 80, 300)$	118 600
EEV		107 240

Hence $\mathrm{EVPI} = 7\,016$ and $\mathrm{VSS} = 1\,150$. Here $\overline{\xi}$ is one of the scenarios, so that $\overline{x}$ is also its wait-and-see decision.

In the course code

`code/LShaped.jl` computes each value, for a problem `farmer` given as data (costs, hence negative values; see section 10 of `LShaped_usage.ipynb`):

```
rp  = lshaped(farmer; optimizer = HiGHS.Optimizer).objective
ws  = wait_and_see(farmer; optimizer = HiGHS.Optimizer)   # S problems
ev  = expected_value_problem(farmer; optimizer = HiGHS.Optimizer)
eev = expected_result(farmer, ev.x; optimizer = HiGHS.Optimizer)
                                    # S subproblems at xbar
evpi(farmer; rp = rp, optimizer = HiGHS.Optimizer)   # RP - WS
vss(farmer; rp = rp, optimizer = HiGHS.Optimizer)    # EEV - RP
```

`wait_and_see`, `expected_result`, `evpi`, and `vss` with a given $\overline{x}$, also apply to problems written as JuMP models; `expected_value_problem` needs scenarios that can be averaged.

Regularization

The L-shaped method suffers from "stability" issues,

- especially during the first iterations when we do not have a good model of $\mathcal{Q}(x)$;
- especially when we start from a good estimate of the solution.

It is therefore useful to regularize the method. Various techniques exist.

Some references on the stabilization of the Benders decomposition:

- Marsten, Hogan, Blankenship (1975);
- Lemaréchal (1975,...);
- Kiwiel (1983,...);
- Neame, Boland, and Ralph (1998).

Regularization of the *L*-Shaped method

- Penalization of the step length: A. Ruszczyński, *A Regularized Decomposition for Minimizing a Sum of Polyhedral Functions*, Mathematical Programming 35, pp. 309–333, 1986.
- Approach in $\|\cdot\|_\infty$: J. T. Linderoth and S. J. Wright, *Implementing a Decomposition Algorithm for Stochastic Programming on a Computational Grid*, Computational Optimization and Applications 24, pp. 207–250, 2003.

Penalization of the step length

At iteration ν, with a^ν the best point found so far (the centre), we solve the program

$$\min_{x,\theta} c^T x + \sum_{k=1}^{C} \theta_k + \frac{1}{2\rho}\|x - a^\nu\|^2.$$

- Large ρ: behaves like standard L-shaped.
- Small ρ: small step size (proximal effect).

We speak of the regularized decomposition method.

Trust-region approach

The principle is also to penalize the step length. At iteration ν, we require $\|x - a^\nu\| \leq \Delta_\nu$, where $\|\cdot\|$ denotes a standard norm (typically 2 or ∞).

- Large Δ_ν: behaves like standard L-shaped.
- Small Δ_ν: restricts movement.

We speak of trust-region regularization.

In both cases, we see the introduction of nonlinear programming concepts.

Summary

Master problem for each regularization approach:

- Multicut L-shaped

$$\min_{x,\theta} \left\{ c^T x + \sum_{k=1}^{C} \theta_k \;\middle|\; (x, \theta_1, \theta_2, \ldots, \theta_C) \in \mathcal{B}_0 \cap \mathcal{B}_1 \right\}.$$

- Trust-region

$$\min_{x,\theta} \left\{ c^T x + \sum_{k=1}^{C} \theta_k \;\middle|\; (x, \theta_1, \theta_2, \ldots, \theta_C) \in \mathcal{B}_0 \cap \mathcal{B}_1,\ \|x - a^\nu\|_\infty \leq \Delta_\nu \right\}.$$

- Regularized decomposition

$$\min_{x,\theta} \left\{ c^T x + \sum_{k=1}^{C} \theta_k + \frac{1}{2\rho}\|x - a^\nu\|_2^2 \;\middle|\; (x, \theta_1, \theta_2, \ldots, \theta_C) \in \mathcal{B}_0 \cap \mathcal{B}_1 \right\}.$$

Regularized decomposition method

Reference: Birge and Louveaux, Section 5.2. Multicut version.

Step 0 Set $\rho = 1$, $r = \nu = 0$, $\kappa(s) = 0$ for all $s = 1, \ldots, S$. Select a feasible solution a^1.

Step 1 Set $\nu := \nu + 1$, and solve the regularized master program

$$\begin{aligned}
\min\ & c^T x + \sum_{s=1}^{S} \theta_s + \frac{1}{2\rho}\|x - a^\nu\|^2, \\
\text{s.t. } & Ax = b, \\
& D_\ell x \geq d_\ell,\ \ell = 1, \ldots, r, \\
& -\beta_{\ell(s)}^T x + \theta_s \geq \alpha_{\ell(s)},\ \ell(s) = 1, \ldots, \kappa(s), \\
& x \geq 0.
\end{aligned}$$

Regularized decomposition method (cont'd)

Step 1 (cont'd) Let (x^ν, θ^ν) be an optimal solution of this program, where $\theta^\nu = (\theta_1^\nu, \ldots, \theta_S^\nu)^T$.
If $\kappa(s) = 0$ for some s, θ_s^ν is ignored in the computation. If $c^T x^\nu + e^T \theta^\nu = c^T a^\nu + \mathcal{Q}(a^\nu)$, stop: a^ν is optimal.

Step 2 If a feasibility cut is generated, set $a^{\nu+1} = a^\nu$ (infeasible null step), and return to Step 1.

Step 3 For $s = 1, \ldots, S$, solve the linear subproblem

$$\begin{aligned} \min_y \ & q_s^T y \\ \text{s.t. } & Wy = h_s - T_s x^\nu, \\ & y \geq 0. \end{aligned}$$

Compute $Q(x^\nu, \xi_s)$.

Regularized decomposition method (cont'd)

Step 3 (cont'd) If

$$\theta_s^\nu < p_s(\pi_s^\nu)^T(h_s - T_s x^\nu),$$

add an optimality cut of the form

$$-\beta_{\ell(s)}^T x + \theta_s \geq \alpha_{\ell(s)}.$$

Set $\kappa(s) := \kappa(s) + 1$.

Step 4 If for all s,

$$\theta_s^\nu \geq p_s(\pi_s^\nu)^T(h_s - T_s x^\nu),$$

set $a^{\nu+1} := x^\nu$ (exact serious step). Return to Step 1.

Regularized decomposition method (cont'd)

Step 5 If

$$c^T x^\nu + \mathcal{Q}(x^\nu) \leq c^T a^\nu + \mathcal{Q}(a^\nu),$$

set $a^{\nu+1} := x^\nu$ (serious approximate step).
Otherwise, set $a^{\nu+1} := a^\nu$.
Return to Step 1.

Immediate extension: allow ρ to vary.

The implementation of the regularized L-shaped method raises various practical questions. An efficient implementation is proposed in Ruszczyński (1986).

Example: regularized L-shaped

Consider again Birge and Louveaux, Section 5.1, Exercise 5, page 200.
Take $a^1 = -0.5$ as starting point. This point corresponds to the solution of the expected value problem, i.e., the problem obtained by replacing ξ by $\mathbb{E}[\xi]$. The second-stage program becomes

$$\begin{aligned} \min\ & \frac{5}{4}y_1 + \frac{1}{7}y_3 + \frac{1}{2}y_4 \\ \text{s.t. } & y_1 - y_2 - y_3 - y_4 = -\frac{1}{2} - x, \\ & y_2 + y_5 = 2, \\ & y_3 + y_6 = 7, \\ & y \geq 0. \end{aligned}$$

Exercise: check with the KKT/primal-dual conditions that $a^1 = -0.5$ is an optimal solution of this program — its solution set is $[-1/2, 3/2]$, and a^1 is the left endpoint.

Example: regularized L-shaped (cont'd)

For $a^1 = -0.5$, the regularized master problem also has solution $x^1 = -0.5$, and the second-stage program of the expected-value problem has the solution $(0, 0, 0, 0, 2, 7)$, while $Q_1(a^1) = 0$, $Q_2(a^1) = 3/4$, so $\mathcal{Q}(a^1) = 3/8$. The regularized approach leads to the following iterations.

Iteration 1 The cuts $\theta_1 \geq 0$, $\theta_2 \geq -3/4x$ are added. $a^2 := a^1$.

Iteration 2 The regularized master program is

$$\min \theta_1 + \theta_2 + \frac{1}{2}(x + 0.5)^2$$
$$\text{s.t. } \theta_1 \geq 0,\ \theta_2 \geq -\frac{3}{4}x,$$

with the solution $x^2 = 0.25$, $\theta_1^2 = 0$, $\theta_2^2 = -3/16$. A new cut $\theta_2 \geq 0$ is added. As $\mathcal{Q}(0.25) = 0 < \mathcal{Q}(a^1)$, $a^3 := 0.25$ (serious approximate step).

Example: regularized *L*-shaped (cont'd)

Iteration 3 The regularized master problem is

$$\min\ \theta_1 + \theta_2 + \frac{1}{2}(x - 0.25)^2$$

$$\text{s.t. } \theta_1 \geq 0,\ \theta_2 \geq -\frac{3}{4}x, \theta_2 \geq 0,$$

with the solution $x^3 = 0.25$, $\theta_1^3 = 0$, $\theta_2^3 = 0$. As $e^T\theta^3 = \mathcal{Q}(a^3)$, a solution is found.

Theorem

If the original problem has a solution, the algorithm stops after a finite number of iterations. Otherwise, it generates a sequence of feasible points $\{a^\nu\}$ *such that* $c^T a^\nu + \mathcal{Q}(a^\nu) \to -\infty$, *as* $\nu \to \infty$.

Proof.

See Birge and Louveaux, Section 5.2.

Regularization using trust-region methods

Basic philosophy of trust-region methods: replace the original problem with some locally valid approximation easier to handle.

Original problem

$$\begin{aligned} \min_{x} \ & c^T x + \mathcal{Q}(x) \\ \text{s.t. } & Ax = b \\ & x \geq 0. \end{aligned}$$

Model

$$\begin{aligned} \min_{x,\theta} \ & m(x,\theta) := c^T x + \theta \\ \text{s.t. } & Ax = b \\ & \text{feasibility cuts} \\ & \text{optimality cuts} \\ & x \geq 0. \end{aligned}$$

Regularization using trust-region methods (cont'd)

- The model $m(\cdot)$ is valid locally only.
- Constrain the candidate iterate to stay in a trust region $\mathcal{T}_\nu$ centered at a^ν, the best point found so far (as in the regularized decomposition):

$$\mathcal{T}_\nu = \{x \mid \|x - a^\nu\|_\nu \leq \Delta_\nu\},$$

  where Δ_ν is the trust-region radius, and $\|\cdot\|_\nu$ is the norm used at iteration ν. ($\mathcal{B}_0$ and $\mathcal{B}_1$ keep their earlier meaning: the constraints of the master program.)
- In the context of linear programming, it is common to use the ∞-norm as

$$\|x - a^\nu\|_\infty \leq \Delta_\nu \iff -\Delta_\nu e \leq x - a^\nu \leq \Delta_\nu e,$$

  where $e = (1, \ldots, 1)^T$.

Regularization using trust-region methods (cont'd)

- The trust-region subproblem at iteration ν is now

$$\begin{aligned} \min_{x,\theta} \; & m(x,\theta) := c^T x + \theta \\ \text{s.t. } & Ax = b \\ & \text{feasibility cuts} \\ & \text{optimality cuts} \\ & \|x - a^\nu\|_\infty \leq \Delta_\nu \\ & x \geq 0. \end{aligned}$$

- We can extend the model to multiple cuts or scenarios partitioned into clusters.
- Only accept the step if the model performs well.

Regularization using trust-region methods (cont'd)

- Let (x^ν, θ^ν) be the solution of the trust-region subproblem, and $\bar{\theta}^\nu$ the model value at the centre a^ν.
- Set $0 < \eta_1 \leq \eta_2 < 1$. For instance, $\eta_1 = \eta_2 = 10^{-4}$.
- Compare the actual reduction to the predicted reduction (τ, not to be confused with the proximal parameter ρ above):

$$\tau = \frac{c^T(a^\nu - x^\nu) + \mathcal{Q}(a^\nu) - \mathcal{Q}(x^\nu)}{m(a^\nu, \bar{\theta}^\nu) - m(x^\nu, \theta^\nu)}$$

- Unsuccessful iteration: if $\tau < \eta_1$, set $a^{\nu+1} = a^\nu$, $\Delta_{\nu+1} \leq \Delta_\nu$, add cuts. Enforce $\Delta_{\nu+1} < \Delta_\nu$ if Δ_ν has been kept the same for several consecutive unsuccessful iterations or if Δ_ν is too large.
- Successful iteration: if $\tau \geq \eta_1$, set $a^{\nu+1} = x^\nu$.
- Very successful iteration: if $\tau \geq \eta_2$, set $\Delta_{\nu+1} = \min\{\Delta_{\max}, \gamma\Delta_\nu\}$, with $\gamma > 1$ (the optimality-cut coefficients β_ℓ are unrelated).

Regularization using trust-region methods: notes

Additional details and considerations in Linderoth and Wright (2003).

- Allow deletion of inactive cuts.
- Investigate parallelization of the method.
- Do not distinguish successful and very successful iterations: $\eta_1 = \eta_2 \in (0, 1/2)$.